

★ PRIMER · UNIT 9

Before the Lecture: No Answer Key This Time

Unit 9 · Unsupervised Learning

Watch this first if centroids, variance and eigenvectors are hazy

Twenty minutes of groundwork — what a centroid is, what a covariance matrix holds, and what it means to keep the most informative angle on your data.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Centroids and covariance are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 9 Sits

The course builds one pipeline; here is this unit's stone

Unit 8

done

Neurons, layers and
backpropagation.

Unit 9

you are here

Structure found without labels.

Unit 10

next

Honest testing and purged CV.

Everything so far had an answer key. From here the machine must find the structure itself.

What Changes Without Labels

With no answer key there is nothing to be right about — only structure to be found, and judged by eye.

Supervised

- You have the answers.
- Score against them.
- Right and wrong are defined.

Unsupervised

- No answers at all.
- Score by structure alone.
- Usefulness replaces accuracy.

This is why unsupervised results need judgement. There is no test set to hide behind.

The Words the Lecture Assumes

Five terms, and both halves of the unit open up

Word	What it means	Here
Centroid	the mean point of a group	the centre of a cluster
Cluster	points closer to each other	stocks that move together
Variance	spread along one direction	how much a factor moves
Covariance	how two series move together	the input to every portfolio
Eigenvector	a direction the matrix prefers	the axis PCA keeps

An eigenvector is just a direction. PCA keeps the ones with the most variance along them.

Two Jobs, One Unit

Clustering groups the rows; PCA shrinks the columns

Task	What it reduces	Method here
Clustering	many rows into few groups	k-means
Dimension reduction	many columns into few	PCA
Denoising	noise in a covariance matrix	random matrix theory

The third is the one that matters most in finance, and unit 11 depends on it entirely.

Why Too Many Features Hurt

In high dimensions everything is far from everything else, and distance stops meaning anything.

In two dimensions

- Near and far are obvious.
- Clusters are visible.
- Distance works.

In five hundred

- All pairs look equally far.
- Clusters dissolve.
- KNN and k-means both fail.

This is the curse of dimensionality, and PCA is the standard answer to it.

One K-Means Step by Hand

Six stocks, two centres — assign each to the nearer one

Stock	Feature	To centre A (2)	To centre B (8)	Goes to
S1	1	1	7	A
S2	3	1	5	A
S3	4	2	4	A
S4	7	5	1	B
S5	9	7	1	B
S6	10	8	2	B

New centre A = $(1+3+4)/3 = 2.67$. New centre B = $(7+9+10)/3 = 8.67$. Then repeat.

Reading That Step

Two moves, alternated until nothing changes

Assign

- Each point to its nearest centre.
- Purely a distance check.
- No labels used anywhere.

Update

- Each centre to its group's mean.
- Centres drift.
- Repeat until they settle.

Different starting centres can land on different answers — always run it more than once

What PCA Keeps

Rotate the data until the first axis carries the most variance

The idea

- Find the widest direction.
- Call it component one.
- Then the next widest.

Why it helps

- Two components often suffice.
- The rest is mostly noise.
- Fewer columns, same story.

In equity returns the first component is usually the market itself, and it is enormous.

Denoising a Correlation Matrix

Three moves — this is the sequence unit 11 relies on

1 · Decompose

find the axes

Break the matrix into directions
and how much variance each
carries.

2 · Compare

to pure noise

Marchenko-Pastur says what
random data would look like.

3 · Flatten

the noise part

Anything inside the noise band
is replaced by its average.

With 500 stocks and 1,000 days, most of the correlation matrix is noise. This finds out how much.

Choosing K, and Its Traps

There is no correct K — only defensible ones

Approach	How it works	Weakness
Elbow	plot error against K	the elbow is often blurred
Silhouette	how tight and separate	slow on large data
Domain sense	how many groups make sense	subjective, and often best

If two methods disagree, that is information — the clusters are probably not real.

Notation You Will Meet

Four symbols carry the whole unit

Symbol	Read it as	It stands for
k	kay	how many clusters you asked for
μ	mu	a centroid — the mean of a group
Σ	sigma	the covariance matrix
λ	lambda	an eigenvalue — variance along one axis

Sigma has two jobs in this course: 'add up' and 'covariance matrix'. Context decides.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>KMeans(n_clusters=3).fit(X)</code>	group the rows
<code>model.cluster_centers_</code>	the centroids it settled on
<code>PCA(n_components=2).fit(X)</code>	find the two widest directions
<code>pca.explained_variance_ratio_</code>	how much each component carries
<code>np.corrcoef(returns.T)</code>	the correlation matrix you will denoise

Print `explained_variance_ratio_` first. In equities the first number is usually startling.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 10

1

"K-means found the real groups"

It found groups. Whether they mean anything is your judgement, not the algorithm's.

2

"PCA components have meanings"

They are directions of variance. Sometimes one is interpretable; usually it is not.

3

"A correlation matrix is data, so it is reliable"

With more assets than days it is mostly noise, and inverting it amplifies exactly that.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1	Points 1, 3, 4 in a cluster. Where is the centroid?	2.67
2	Which unit's portfolio maths depends on denoising?	unit 11
3	Why does distance stop working in 500 dimensions?	all pairs far

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It clusters, by hand and in code

K-means on six stocks, the elbow for choosing K, and clustering days into regimes.

2

It reduces the columns

PCA gently, then the 2×2 case worked in full, then random matrix theory for denoising.

3

It ends in code

The problem set runs k-means by hand, computes an eigenvalue, and cleans a matrix.

Next: Unit 9 lecture — Unsupervised Learning